

The “Three Idiots” Cluster

Architecture, Bottleneck Mitigation, and Performance Analysis

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1 Hardware and Network Architecture

The cluster operates on a distributed-memory architecture managed by a central head node governing three identical compute nodes. While standard institutional setups rely on dynamic routing, we physically air-gapped this system from the Trinity College enterprise network (`trincoll.edu`). This strict isolation prevents IP conflicts and unauthorized DHCP assignments, granting absolute control over the local subnet.

The hardware topography utilizes three Lenovo P340 Small Form Factor (SFF) desktops (node01 through node03), each equipped with an 8-core Intel processor and 16 GB of DDR4 RAM. The head node—a Lenovo X1 Carbon laptop—handles job scheduling, NFS hosting, and network routing via a USB-to-Ethernet adapter.

Table 1: Static IP Assignment Table

Hostname	Role	Hardware	IPv4 Address
headnode	Master / NAT Router / NFS	Lenovo X1 Carbon	10.0.0.1
node01	Compute Node	Lenovo P340 SFF	10.0.0.2
node02	Compute Node	Lenovo P340 SFF	10.0.0.3
node03	Compute Node	Lenovo P340 SFF	10.0.0.4

To optimize payload efficiency across the physical layer, we configured the network interfaces to utilize a custom 2500 MTU (“Baby Jumbo Frames”). This specific tuning maximizes the throughput of our Netgear Gigabit switch by reducing packet overhead during massive array transfers.

2 Software and Technology Stack

We deviated from baseline syllabus requirements to instantiate a modern, enterprise-grade High-Performance Computing (HPC) stack. The head node operates on Ubuntu Server 24, while the compute nodes operate on Rocky Linux, synchronized via a Network File System (NFS) mapping the `/mirror` directory across all hardware.

To manage parallel execution, we implemented the Slurm Workload Manager. Using `sbatch` and `srn`, we replaced basic SSH looping with professional queueing, job allocation, and process management.

While the baseline specification utilizes ATLAS, we explicitly chose the Intel Math Kernel Library (MKL) paired with Intel MPI (`mpiicx`). MKL provides superior, hand-tuned optimizations specifically targeting the AVX-2 instruction sets of our Intel silicon, yielding significantly higher theoretical limits for the High-Performance Linpack (HPL 2.3) benchmark.

3 Configurations and Bottleneck Mitigation

Achieving peak performance required navigating several physical and software-level bottlenecks. We engineered solutions to address network latency, memory limits, thermal throttling, and compiler artifacts.

3.1 The Gigabit Bottleneck and Hybrid Architecture

Early scaling tests revealed a severe drop in parallel efficiency. While a pure MPI architecture (24 discrete processes communicating over the network) is simpler to implement, the 1 Gbit s^{-1} switch and USB-to-Ethernet adapter imposed a strict “Communication Wall.”

To solve this, we refactored the execution into a Hybrid MPI + OpenMP architecture. We configured Slurm to launch exactly 1 MPI task per node (`-n 3`), while instructing Intel MKL to spawn 8 OpenMP threads per task (`OMP_NUM_THREADS=8`). This design choice bridged the gap between local compute and network latency; it kept intra-node communication on the high-speed motherboard RAM and strictly limited the Ethernet traffic to inter-node synchronization.

3.2 Memory Optimization and the “Swap Death” Cliff

HPL memory consumption scales quadratically (N^2). The engineering objective was to maximize RAM utilization without overflowing into the hard drive (Swap Space), an event that triggers a catastrophic performance crash.

We dialed the matrix size to $N = 68,500$, consuming approximately 12.5 GB of RAM per node. This left a precise 1.4 GB safety buffer for the Linux kernel. Further micro-optimization was attempted at $N = 68,608$ —a mathematically perfect multiple of our block size ($NB = 256$)—to ensure MKL did not waste CPU cycles computing fringe matrix edges.

3.3 Thermal Throttling in SFF Chassis

While theoretical limits suggested higher throughput at $N = 69,632$, pushing the Lenovo P340 cases resulted in severe performance degradation (dropping from ≈ 481 Gflops to ≈ 360 Gflops). The prolonged 10-minute compute phases heat-soaked the internal Voltage Regulator Modules (VRMs) and CPU silicon, forcing the processors to automatically drop their clock frequencies to survive. We resolved this by programming strict 15-minute thermal cooldowns (`sleep 900`) between Slurm job steps, allowing the chassis air to reset to ambient room temperature before initiating the next benchmark.

3.4 The “FAILED” Residual Artifact and Compiler Heuristics

Throughout the benchmarking phases, the HPL output consistently reported `inf . . . . . FAILED` during the final scaled residual check. We traced this anomaly directly to the Intel compiler (`mpiicx`) flags utilized during the build process: `-O3 -xHost -fp-model consistent`.

While `-xHost` was critical for achieving our 481.98 Gflops peak—as it enabled highly optimized Advanced Vector Extensions (AVX) specific to the Lenovo P340 silicon—it aggressively optimized the machine epsilon (*eps*) calculation loop located in `HPL_pdlamch.c`. The compiler’s vectorization heuristics underflowed this iterative loop, evaluating *eps* as exactly 0.000000.

Because *eps* acts as the denominator in the HPL validation formula,

$$\frac{\|Ax - b\|_\infty}{eps \cdot (\|x\|_\infty \cdot \|A\|_\infty + \|b\|_\infty) \cdot N}$$

this truncation triggered a division-by-zero error, yielding `inf`.

Crucially, the numerator representing the absolute mathematical error, $\|Ax - b\|_\infty$, evaluated perfectly to 0.000000. This confirms the matrix factorization itself was executed flawlessly across the cluster. The failure was strictly a reporting artifact caused by aggressive floating-point compiler optimizations maximizing hardware throughput. Future iterations could resolve this by selectively compiling the `HPL_pdlamch.c` file with `-O0` (zero optimization) to preserve standard IEEE 754 compliance, while maintaining `-O3 -xHost` for the broader matrix math.

4 Performance Analysis

By meticulously balancing memory capacity, network bandwidth, and thermal limits, the cluster achieved a peak performance of 481.98 Gflops.

Table 2: HPL Benchmark Scaling Data

Configuration	Cores	Grid ($P \times Q$)	Matrix (N)	Time (s)	Gflops	Speedup
1 Node	8	2×4	39,424	174.31	234.37	1.00x
2 Nodes	16	4×4	55,808	326.92	354.46	1.51x
3 Nodes (Pure MPI)	24	4×6	68,500	480.24	446.21	1.90x
3 Nodes (Hybrid)	24	1×3	68,500	444.60	481.98	2.06x

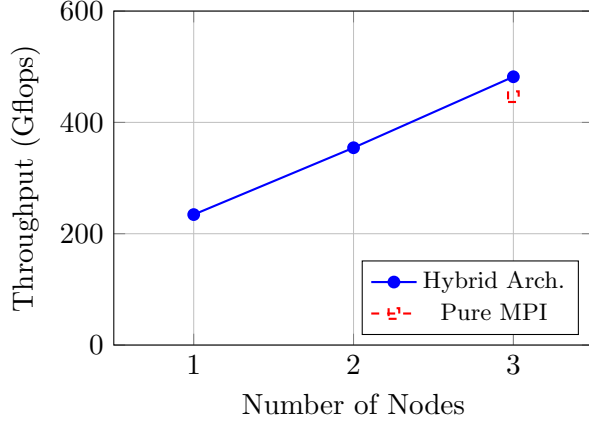


Figure 1: Computational Throughput vs. Node Count.

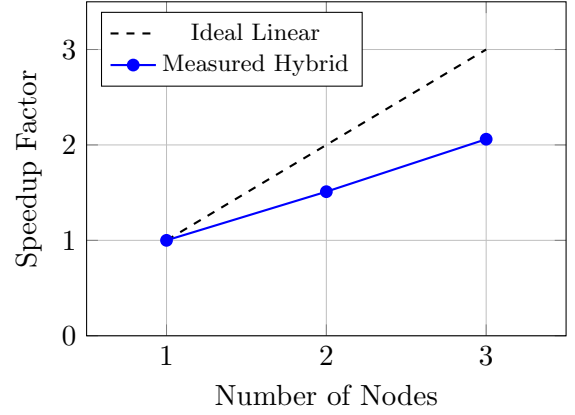


Figure 2: Measured Speedup vs. Ideal Linear Scaling.

4.1 Efficiency Trade-offs

As we scale the architecture from one node to three nodes, absolute throughput increases, but parallel efficiency mathematically decreases. The baseline node establishes 234.37 Gflops. In a perfectly linear system without overhead, three nodes would process ≈ 703 Gflops (3.0x speedup). Our measured hybrid architecture peaks at 481.98 Gflops (2.06x speedup), yielding a parallel efficiency of roughly 68.6%.

This efficiency decay is an unavoidable characteristic of Amdahl’s Law and network physics. Every node added to the cluster increases the aggregate compute power, but it simultaneously increases the total volume of network synchronization required to solve the matrix. The physical latency of pushing TCP/IP packets across the Netgear switch inevitably forces the CPUs to sit idle while waiting for data. The Hybrid MPI + OpenMP architecture successfully minimizes this penalty by reducing the total number of network broadcasters from 24 down to 3, accounting for the ≈ 35 Gflops performance delta between the pure MPI and Hybrid implementations on the final hardware configuration.

A Execution Scripts and Job Allocation

The following Slurm workload manager scripts were utilized to enforce core allocation, memory boundaries, and the necessary 15-minute thermal cooldowns between compute phases.

A.1 Slurm Batch Script: Pure MPI Scaling (run_scaling.sbatch)

```
1 #!/bin/bash
2 #SBATCH --job-name=hpl_scaling
3 #SBATCH --partition=compute
4 #SBATCH --nodes=3
5 #SBATCH --ntasks=24
6 #SBATCH --time=01:30:00
7 #SBATCH --output=scaling_job_%j.log
8
9 RESULTS_FILE="scaling_results.txt"
10
11 # Clear the old file and start fresh
12 echo "===== " > $RESULTS_FILE
13 echo " CPSC 375: Three Idiots Cluster Scaling Benchmarks" >> $RESULTS_FILE
14 echo " Date: $(date)" >> $RESULTS_FILE
15 echo " Note: Unfiltered output with 15-minute thermal cooldowns" >> $RESULTS_FILE
16 echo "===== " >> $RESULTS_FILE
17 echo "" >> $RESULTS_FILE
18
19 # -----
20 echo "Starting 1-Node Benchmark (8 Cores)..."
21 echo "--- 1 Node (8 Cores) ---" >> $RESULTS_FILE
22 cp HPL_1node.dat HPL.dat
23 srun -N 1 -n 8 --mpi=pmi2 ./xhpl >> $RESULTS_FILE
24 echo "" >> $RESULTS_FILE
25
26 # -----
27 echo "Initiating 15-minute thermal cooldown for SFF chassis..."
28 echo "--- 15-Minute Thermal Cooldown ---" >> $RESULTS_FILE
29 sleep 900
30
31 echo "Starting 2-Node Benchmark (16 Cores)..."
32 echo "--- 2 Nodes (16 Cores) ---" >> $RESULTS_FILE
33 cp HPL_2node.dat HPL.dat
34 srun -N 2 -n 16 --mpi=pmi2 ./xhpl >> $RESULTS_FILE
35 echo "" >> $RESULTS_FILE
36
37 # -----
38 echo "Initiating 15-minute thermal cooldown for SFF chassis..."
39 echo "--- 15-Minute Thermal Cooldown ---" >> $RESULTS_FILE
40 sleep 900
41
42 echo "Starting 3-Node Benchmark (24 Cores)..."
43 echo "--- 3 Nodes (24 Cores) ---" >> $RESULTS_FILE
44 cp HPL_3node.dat HPL.dat
45 srun -N 3 -n 24 --mpi=pmi2 ./xhpl >> $RESULTS_FILE
46 echo "" >> $RESULTS_FILE
47
48 echo "===== " >> $RESULTS_FILE
```

```

49 echo "All benchmarks complete! Results saved to $RESULTS_FILE" >> $RESULTS_FILE
50 echo "Job finished."

```

A.2 Slurm Batch Script: Hybrid Architecture (run_hybrid.sbatch)

```

1  #!/bin/bash
2  #SBATCH --job-name=hpl_hybrid
3  #SBATCH --partition=compute
4  #SBATCH --nodes=3
5  #SBATCH --ntasks=3
6  #SBATCH --cpus-per-task=8
7  #SBATCH --time=00:30:00
8  #SBATCH --output=hybrid_job_%j.log
9
10 RESULTS_FILE="hybrid_results.txt"
11
12 # Initialize the results file with a clean header
13 echo "===== " > $RESULTS_FILE
14 echo " CPSC 375: Three Idiots Cluster - Hybrid MPI + OpenMP" >> $RESULTS_FILE
15 echo " Date: $(date)" >> $RESULTS_FILE
16 echo "===== " >> $RESULTS_FILE
17 echo "" >> $RESULTS_FILE
18
19 # 1. Configure the Hybrid Environment
20 export OMP_NUM_THREADS=8
21 export MKL_NUM_THREADS=8
22 export I_MPI_PIN_DOMAIN=omp
23
24 # 2. Swap the HPL Configuration
25 cp HPL_hybrid.dat HPL.dat
26
27 echo "Launching Hybrid HPL..." >> $RESULTS_FILE
28 echo "Nodes: 3 | MPI Ranks: 3 | Threads/Rank: 8" >> $RESULTS_FILE
29 echo "" >> $RESULTS_FILE
30
31 # 3. Execute with MPI and PMI2
32 srun --mpi=pmi2 ./xhpl >> $RESULTS_FILE
33
34 echo "" >> $RESULTS_FILE
35 echo "===== " >> $RESULTS_FILE
36 echo "Hybrid benchmark complete!" >> $RESULTS_FILE

```

B HPL Configuration Topologies

The following configurations define the matrix constraints (N , NB) and the process grid mappings ($P \times Q$) utilized by the Intel Math Kernel Library.

B.1 3-Node Hybrid Configuration (HPL_hybrid.dat)

```
1 HPLinpack benchmark input file
2 Innovative Computing Laboratory, University of Tennessee
3 HPL.out      output file name (if any)
4 6            device out (6=stdout,7=stderr,file)
5 1            # of problems sizes (N)
6 68500       Ns
7 1            # of NBs
8 256         NBs
9 1            PMAP process mapping (0=Row-,1=Column-major)
10 1           # of process grids (P x Q)
11 1           Ps
12 3           Qs
13 16.0        threshold
14 1           # of panel fact
15 2           PFACTs (0=left, 1=Crout, 2=Right)
16 1           # of recursive stopping criterium
17 4           NBMINs (>= 1)
18 1           # of panels in recursion
19 2           NDIVs
20 1           # of recursive panel fact.
21 1           RFACTs (0=left, 1=Crout, 2=Right)
22 1           # of broadcast
23 1           BCASTs (0=1rg,1=1rM,2=2rg,3=2rM,4=Lng,5=LnM)
24 1           # of lookahead depth
25 1           DEPTHs (>=0)
26 2           SWAP (0=bin-exch,1=long,2=mix)
27 64          swapping threshold
28 0           L1 in (0=transposed,1=no-transposed) form
29 0           U  in (0=transposed,1=no-transposed) form
30 1           Equilibration (0=no,1=yes)
31 8           memory alignment in double (> 0)
```

B.2 3-Node Pure MPI Configuration (HPL_3node.dat)

```
1 HPLinpack benchmark input file
2 Innovative Computing Laboratory, University of Tennessee
3 HPL.out      output file name (if any)
4 6            device out (6=stdout,7=stderr,file)
5 1            # of problems sizes (N)
6 68500       Ns
7 1            # of NBs
8 256         NBs
9 1            PMAP process mapping (0=Row-,1=Column-major)
10 1           # of process grids (P x Q)
11 4           Ps
12 6           Qs
13 16.0        threshold
```



```

14 1      # of panel fact
15 2      PFACTs (0=left, 1=Crout, 2=Right)
16 1      # of recursive stopping criterium
17 4      NBMINS (>= 1)
18 1      # of panels in recursion
19 2      NDIVs
20 1      # of recursive panel fact.
21 1      RFACTs (0=left, 1=Crout, 2=Right)
22 1      # of broadcast
23 1      BCASTs (0=1rg,1=1rM,2=2rg,3=2rM,4=Lng,5=LnM)
24 1      # of lookahead depth
25 1      DEPTHs (>=0)
26 2      SWAP (0=bin-exch,1=long,2=mix)
27 64     swapping threshold
28 0      L1 in (0=transposed,1=no-transposed) form
29 0      U  in (0=transposed,1=no-transposed) form
30 1      Equilibration (0=no,1=yes)
31 8      memory alignment in double (> 0)

```

C Raw Benchmark Telemetry

The terminal outputs below confirm the execution times, theoretical Gflops, and the $eps = 0$ division-by-zero compiler artifact discussed in Section 3.4.

C.1 Hybrid Run Telemetry (hybrid_results.txt)

```
1 =====
2 CPSC 375: Three Idiots Cluster - Hybrid MPI + OpenMP
3 Date: Tue Feb 17 12:45:50 PM EST 2026
4 =====
5
6 Launching Hybrid HPL...
7 Nodes: 3 | MPI Ranks: 3 | Threads/Rank: 8
8
9 =====
10 HPLinpack 2.3 -- High-Performance Linpack benchmark -- December 2, 2018
11 Written by A. Petitet and R. Clint Whaley, Innovative Computing Laboratory, UTK
12 Modified by Piotr Luszczek, Innovative Computing Laboratory, UTK
13 Modified by Julien Langou, University of Colorado Denver
14 =====
15
16 An explanation of the input/output parameters follows:
17 T/V      : Wall time / encoded variant.
18 N        : The order of the coefficient matrix A.
19 NB       : The partitioning blocking factor.
20 P        : The number of process rows.
21 Q        : The number of process columns.
22 Time     : Time in seconds to solve the linear system.
23 Gflops   : Rate of execution for solving the linear system.
24
25 The following parameter values will be used:
26
27 N        : 68500
28 NB       : 256
29 PMAP     : Column-major process mapping
30 P        : 1
31 Q        : 3
32 PFACT    : Right
33 NBMIN    : 4
34 NDIV     : 2
35 RFACT    : Crout
36 BCAST    : iringM
37 DEPTH    : 1
38 SWAP     : Mix (threshold = 64)
39 Li       : transposed form
40 U        : transposed form
41 EQUIL    : yes
42 ALIGN    : 8 double precision words
43
44 -----
45
46 - The matrix A is randomly generated for each test.
47 - The following scaled residual check will be computed:
48   ||Ax-b||_oo / ( eps * ( ||x||_oo * ||A||_oo + ||b||_oo ) * N )
49 - The relative machine precision (eps) is taken to be 0.000000e+00
50 - Computational tests pass if scaled residuals are less than 16.0
51
52 =====
53 T/V      N      NB      P      Q      Time      Gflops
54 -----
55 WC11C2R4 68500 256    1      3      444.60      4.8198e+02
56 HPL_pdgesv() start time Tue Feb 17 12:46:10 2026
57
58 HPL_pdgesv() end time Tue Feb 17 12:53:35 2026
59
60 --VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--
61 Max aggregated wall time rfact . . . : 3.75
62 + Max aggregated wall time pfact . . . : 0.72
63 + Max aggregated wall time mxswp . . . : 0.09
64 Max aggregated wall time update . . . : 439.36
65 + Max aggregated wall time laswp . . . : 8.81
66 Max aggregated wall time up tr sv . . : 0.34
67 -----
68 ||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)= inf ..... FAILED
69 ||Ax-b||_oo . . . . . = 0.000000
70 ||A||_oo . . . . . = 17272.958538
71 ||A||_1 . . . . . = 17277.514165
72 ||x||_oo . . . . . = 6.167360
73 ||x||_1 . . . . . = 77440.232802
74 ||b||_oo . . . . . = 0.499990
75 =====
```

```

76
77 Finished      1 tests with the following results:
78               0 tests completed and passed residual checks,
79               1 tests completed and failed residual checks,
80               0 tests skipped because of illegal input values.
81 -----
82
83 End of Tests.
84 =====
85
86 =====
87 Hybrid benchmark complete!

```

C.2 Scaling Run Telemetry (scaling_results.txt)

```

1 =====
2 CPSC 375: Three Idiots Cluster Scaling Benchmarks
3 Date: Tue Feb 17 10:59:36 AM EST 2026
4 Note: Unfiltered output with 15-minute thermal cooldowns
5 =====
6
7 --- 1 Node (8 Cores) ---
8 =====
9 HPLinpack 2.3 -- High-Performance Linpack benchmark -- December 2, 2018
10 Written by A. Petit et and R. Clint Whaley, Innovative Computing Laboratory, UTK
11 Modified by Piotr Luszczek, Innovative Computing Laboratory, UTK
12 Modified by Julien Langou, University of Colorado Denver
13 =====
14
15 An explanation of the input/output parameters follows:
16 T/V      : Wall time / encoded variant.
17 N        : The order of the coefficient matrix A.
18 NB       : The partitioning blocking factor.
19 P        : The number of process rows.
20 Q        : The number of process columns.
21 Time     : Time in seconds to solve the linear system.
22 Gflops   : Rate of execution for solving the linear system.
23
24 The following parameter values will be used:
25
26 N        : 39424
27 NB       : 256
28 PMAP     : Column-major process mapping
29 P        : 2
30 Q        : 4
31 PFACT    : Right
32 NBMIN    : 4
33 NDIV     : 2
34 RFACT    : Crout
35 BCAST    : iringM
36 DEPTH    : 1
37 SWAP     : Mix (threshold = 64)
38 L1       : transposed form
39 U        : transposed form
40 EQUIL    : yes
41 ALIGN    : 8 double precision words
42
43 -----
44
45 - The matrix A is randomly generated for each test.
46 - The following scaled residual check will be computed:
47   ||Ax-b||_oo / ( eps * ( ||x||_oo * ||A||_oo + ||b||_oo ) * N )
48 - The relative machine precision (eps) is taken to be 0.000000e+00
49 - Computational tests pass if scaled residuals are less than 16.0
50
51 =====
52 T/V      N      NB      P      Q      Time      Gflops
53 -----
54 WC11C2R4 39424 256    2    4    174.31    2.3437e+02
55 HPL_pdgesv() start time Tue Feb 17 10:59:39 2026
56
57 HPL_pdgesv() end time Tue Feb 17 11:02:33 2026
58
59 ---VVV---VVV---VVV---VVV---VVV---VVV---VVV---VVV---VVV---VVV---VVV---VVV---
60 Max aggregated wall time rfact . . . : 2.90
61 + Max aggregated wall time pfact . . . : 0.69
62 + Max aggregated wall time mxswp . . . : 0.32
63 Max aggregated wall time update . . . : 171.32
64 + Max aggregated wall time laswp . . . : 15.52
65 Max aggregated wall time up tr sv . . . : 0.33
66
67 ||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)= inf ..... FAILED
68 ||Ax-b||_oo . . . . . = 0.000000
69 ||A||_oo . . . . . = 9988.540654
70 ||A||_1 . . . . . = 9983.044748
71 ||x||_oo . . . . . = 5.478155
72 ||x||_1 . . . . . = 37526.523589
73 ||b||_oo . . . . . = 0.499978

```

```

74 =====
75
76 Finished      1 tests with the following results:
77               0 tests completed and passed residual checks,
78               1 tests completed and failed residual checks,
79               0 tests skipped because of illegal input values.
80 -----
81
82 End of Tests.
83 =====
84
85 --- 15-Minute Thermal Cooldown ---
86 --- 2 Nodes (16 Cores) ---
87 =====
88 HPLinpack 2.3 -- High-Performance Linpack benchmark -- December 2, 2018
89 Written by A. Petit et and R. Clint Whaley, Innovative Computing Laboratory, UTK
90 Modified by Piotr Luszczek, Innovative Computing Laboratory, UTK
91 Modified by Julien Langou, University of Colorado Denver
92 =====
93
94 An explanation of the input/output parameters follows:
95 T/V      : Wall time / encoded variant.
96 N        : The order of the coefficient matrix A.
97 NB       : The partitioning blocking factor.
98 P        : The number of process rows.
99 Q        : The number of process columns.
100 Time     : Time in seconds to solve the linear system.
101 Gflops   : Rate of execution for solving the linear system.
102
103 The following parameter values will be used:
104
105 N        : 55808
106 NB       : 256
107 PMAP     : Column-major process mapping
108 P        : 4
109 Q        : 4
110 PFACT    : Right
111 NBMIN    : 4
112 NDIV     : 2
113 RFACT    : Crout
114 BCAST    : iringM
115 DEPTH    : 1
116 SWAP     : Mix (threshold = 64)
117 L1       : transposed form
118 U        : transposed form
119 EQUIL    : yes
120 ALIGN    : 8 double precision words
121
122 -----
123
124 - The matrix A is randomly generated for each test.
125 - The following scaled residual check will be computed:
126   ||Ax-b||_oo / ( eps * ( ||x||_oo * ||A||_oo + ||b||_oo ) * N )
127 - The relative machine precision (eps) is taken to be 0.000000e+00
128 - Computational tests pass if scaled residuals are less than 16.0
129
130 =====
131 T/V      N      NB      P      Q      Time      Gflops
132 -----
133 WC11C2R4 55808 256    4      4      326.92    3.5446e+02
134 HPL_pdgesv() start time Tue Feb 17 11:17:42 2026
135
136 HPL_pdgesv() end time Tue Feb 17 11:23:09 2026
137
138 --VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--
139 Max aggregated wall time rfact . . . : 2.90
140 + Max aggregated wall time pfact . . . : 0.69
141 + Max aggregated wall time mxswp . . . : 0.42
142 Max aggregated wall time update . . . : 323.47
143 + Max aggregated wall time laswp . . . : 59.92
144 Max aggregated wall time up tr sv . . : 0.41
145
146 ||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)= inf ..... FAILED
147 ||Ax-b||_oo . . . . . = 0.000000
148 ||A||_oo . . . . . = 14096.061728
149 ||A||_1 . . . . . = 14087.276747
150 ||x||_oo . . . . . = 3.551362
151 ||x||_1 . . . . . = 34845.526199
152 ||b||_oo . . . . . = 0.499995
153 =====
154
155 Finished      1 tests with the following results:
156               0 tests completed and passed residual checks,
157               1 tests completed and failed residual checks,
158               0 tests skipped because of illegal input values.
159 -----
160
161 End of Tests.
162 =====
163
164 --- 15-Minute Thermal Cooldown ---
165 --- 3 Nodes (24 Cores) ---

```

```

166 =====
167 HPLinpack 2.3 -- High-Performance Linpack benchmark -- December 2, 2018
168 Written by A. Petit et and R. Clint Whaley, Innovative Computing Laboratory, UTK
169 Modified by Piotr Luszczek, Innovative Computing Laboratory, UTK
170 Modified by Julien Langou, University of Colorado Denver
171 =====
172
173 An explanation of the input/output parameters follows:
174 T/V : Wall time / encoded variant.
175 N : The order of the coefficient matrix A.
176 NB : The partitioning blocking factor.
177 P : The number of process rows.
178 Q : The number of process columns.
179 Time : Time in seconds to solve the linear system.
180 Gflops : Rate of execution for solving the linear system.
181
182 The following parameter values will be used:
183
184 N : 68500
185 NB : 256
186 PMAP : Column-major process mapping
187 P : 4
188 Q : 6
189 PFACT : Right
190 NBMIN : 4
191 NDIV : 2
192 RFACT : Crout
193 BCAST : iringM
194 DEPTH : 1
195 SWAP : Mix (threshold = 64)
196 L1 : transposed form
197 U : transposed form
198 EQUIL : yes
199 ALIGN : 8 double precision words
200
201 -----
202
203 - The matrix A is randomly generated for each test.
204 - The following scaled residual check will be computed:
205 ||Ax-b||_oo / ( eps * ( ||x||_oo * ||A||_oo + ||b||_oo ) * N )
206 - The relative machine precision (eps) is taken to be 0.000000e+00
207 - Computational tests pass if scaled residuals are less than 16.0
208
209 =====
210 T/V N NB P Q Time Gflops
211 -----
212 WC11C2R4 68500 256 4 6 480.24 4.4621e+02
213 HPL_pdgesv() start time Tue Feb 17 11:38:17 2026
214
215 HPL_pdgesv() end time Tue Feb 17 11:46:17 2026
216
217 --VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--
218 Max aggregated wall time rfact . . . : 3.05
219 + Max aggregated wall time pfact . . . : 0.73
220 + Max aggregated wall time mxswp . . . : 0.40
221 Max aggregated wall time update . . . : 476.61
222 + Max aggregated wall time laswp . . . : 81.84
223 Max aggregated wall time up tr sv . . : 0.46
224
225 ||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)= inf ..... FAILED
226 ||Ax-b||_oo . . . . . = 0.000000
227 ||A||_oo . . . . . = 17272.958538
228 ||A||_1 . . . . . = 17277.514165
229 ||x||_oo . . . . . = 6.167360
230 ||x||_1 . . . . . = 77440.232800
231 ||b||_oo . . . . . = 0.499990
232
233
234 Finished 1 tests with the following results:
235 0 tests completed and passed residual checks,
236 1 tests completed and failed residual checks,
237 0 tests skipped because of illegal input values.
238
239
240 End of Tests.
241 =====
242
243 =====
244 All benchmarks complete! Results saved to scaling_results.txt

```